367-2-5881 Physiology of the Motor System 2027-1

Meetings	Tuesday, 10:00	Format	Six lectures + six discussions
Text	Kandel, 6th ed. (2021)	Resources	Moodle; public links to follow

Course description

A survey of the different brain areas associated with motor control and motor learning. These include frontal and parietal cortex, basal ganglia, cerebellum, red nucleus, superior colliculus, and spinal cord. We will discuss the oculomotor system and the somatic motor system and their inter-relationships. The role of afferent feedback and efferent copy in different motor areas will also be covered. All of this will also be discussed in the context of plasticity and motor learning.

Overview

I will start the course with 6 lectures on the motor system. I will clarify the different motor systems that are studied by neurophysiologists, how these motor systems are connected to different areas of the brain, and then address the different motor areas, their anatomical interconnections, and the 'textbook' understanding of their function. These lectures will be supplemented with reading from a classic neurophysiology textbook (Kandel, Schwartz, and Jessel, Principles of Neural Science). For the remainder of the semester, we will read papers that cover different topics in the neurophysiology of the motor system. For each meeting, we will go over one paper that will be read by the entire class. In the lecture, I will provide the background for a deeper understanding of the paper but the students will be responsible for explaining the methods and the results of the paper itself in an open discussion.

Reading before class

Assigned readings are to be completed **before the corresponding class**. For textbook lectures, read the indicated Kandel chapters in advance. For research-paper discussions, read the assigned paper before class and come prepared to explain and discuss its methods and results. Research-paper selections will be announced in advance. You are free to use any AI assistance that feels appropriate to help you understand the paper, but I recommend that you also spend some time reading the actual words with your own eyes and looking at the figures.

Written assignments

Weekly short written assignments in English are required. They ensure that students complete the readings and practice written expression. Assignments begin at 150 words and become progressively longer, reaching a length of 500 words. Use of AI is permitted, and AI will also be used in grading. Grading details remain to be published.

Final quiz

A paper-based, multiple-choice, closed-book quiz, designed for approximately 90 minutes. It will take place separately during the university examination period, beginning 17 January 2027.

Grading

Attendance — 10%
Attendance contributes ten percent of the base course grade.

Writing — 35%
The weekly written assignments together contribute thirty-five percent.

Final quiz — 55%
The paper-based, multiple-choice, closed-book final quiz contributes fifty-five percent of the course grade.

Course schedule

Week 1 October 20–26	**L01 · Lecture** — Principles of sensorimotor control *Reading: Kandel Ch. 30*
Week 2 October 27–November 2	**Knesset election day Tuesday, Oct. 27 (no classes)** *No regular class meeting.*
Week 3 November 3–9	**L02 · Lecture** — Voluntary movement *Reading: Kandel Ch. 34 · Writing due: 150 words*

Week 4 November 10–16	**L03 · Lecture** — Eye movements and posture *Reading: Kandel Ch. 35 · Writing due: 150 words*
Week 5 November 17–23	**L04 · Lecture** — Cerebellum and basal ganglia *Reading: Kandel Chs. 37–38 · Writing due: 150 words*
Week 6 November 24–30	**L05 · Lecture** — Spinal cord *Reading: Kandel Chs. 31–32 · Writing due: 250 words*
Week 7 December 1–7	**L06 · Lecture** — Locomotion, posture and balance *Reading: Kandel Chs. 33, 36 · Writing due: 250 words*
Week 8 December 8–14	**D01 · Discussion** — Neural manifolds *Reading: To be announced · Writing due: none*
Week 9 December 15–21	**D02 · Discussion** — Freely moving animals *Reading: To be announced · Writing due: 350 words*
Week 10 December 22–28	**D03 · Discussion** — Vigor *Reading: To be announced · Writing due: none*
Week 11 December 29–January 4	**D04 · Discussion** — Cerebellar function *Reading: To be announced · Writing due: 500 words*
Week 12 January 5–11	**D05 · Discussion** — Basal ganglia *Reading: To be announced · Writing due: none*
Week 13 January 12–12	**D06 · Discussion** — Brain-computer interface *Reading: To be announced · Writing due: 500 words*

Assigned readings

Textbook: Kandel et al., *Principles of Neural Science*, 6th ed. (McGraw-Hill, 2021).

Textbook lectures

L1	Ch. 30	The different motor systems. A summary of the different motor systems that are used and the different approaches to each one
L2	Ch. 34	Motor cortical areas and voluntary movement. Primary motor cortex, premotor cortex, posterior parietal cortex, and anterior cingulate cortex
L3	Ch. 35	Control of eye movements: brainstem nuclei, types of eye movements, eye movement command centers
L4	Chs. 37–38	Motor control loops: cerebellum, basal ganglia, and the thalamus
L5	Chs. 31–32	Spinal cord and locomotion and central pattern generators
L6	Chs. 33, 36	Locomotion, posture and balance

Research-paper discussions

The paper selections for D01–D06 are **to be announced**. The discussion topics and dates appear in the course schedule. Each paper must be read **before its discussion meeting**.